IoT Beehive Monitoring System

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***Abstract* — The paper presents a system that represents an innovative approach to the management of beehives through the integration of modern technology. The system is designed to monitor various critical parameters within the hive and provide real-time feedback on its health and productivity. The novelty of this project lies in the use of a modular and open-source platform that utilizes wireless communication technologies for data acquisition and transmission and leverages the well-documented ESP32 development platform. This setup allows for a scalable and customizable solution that can be adapted to the specific needs of individual beekeepers. The main parameters to be monitored are temperature, humidity, hive weight, CO2 concentration and general hive monitoring. For use in remote areas, the system is equipped with a solar panel to utilize renewable energy, coupled with a battery management system that ensures the sustainability and reliability of the system. The main achievement of this project is the creation of a user-friendly and efficient monitoring platform that can significantly improve the users' experience, guiding them to make better management decisions for the hive. In summary, the Beehive Monitoring System is a practical and innovative solution for small and medium sized beekeepers that combines modern technology with the basic requirements of hive management, providing a powerful tool to improve the sustainability of bee colonies.**

Keywords— Hive Monitoring, Beekeeping Technology, Affordable, Open source, IoT.

I. Introduction

One of the biggest challenges for beekeepers is the limited access to modern technology for monitoring and managing hives. A smart monitoring system provides beekeepers with real-time data and information on the condition of the hives and the activity of the bees, so that constant physical presence is no longer necessary.

This project was developed according to beekepers needs and aims to set an example for this market segment. Thus, the complete monitoring solution includes an online platform that can be accessed from any device and allows beekeepers to monitor their hives. The aim was to create a node for monitoring beehives, which can then be integrated into a network so that the system can subsequently be expanded by adding new identical modules. Monitoring is carried out using data acquisition modules based on ESP32 for data processing and transmission. The system is powered by a small solar panel and a Li-Ion battery in the form of a 18650 cell. The process begins by recognizing and checking the connected sensors. After these initial steps, the connection to the Firebase services is established to recognize the hive and the user and establish the connection. Once all operational requirements are met, the system sends the data provided by the data acquisition system to the Firebase real-time database at set intervals. Compared to existing solutions [1][2], this system offers a more cost-effective and accessible alternative with the added benefit of modularity.

II. The Monitoring System

A. Hardware

The block diagram of the proposed system is shown in Figure 1.

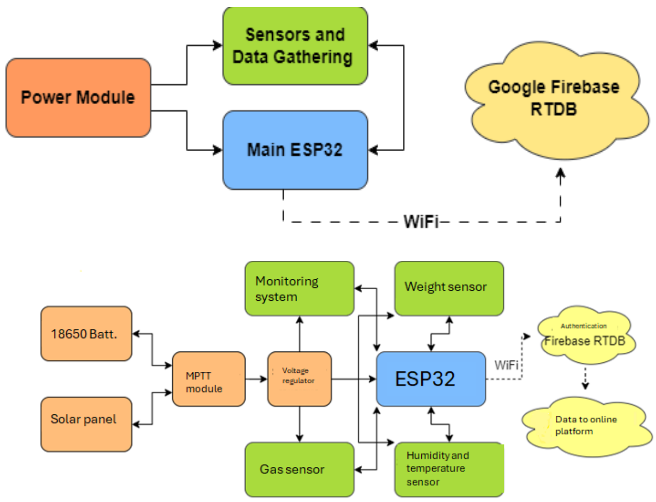

Fig. 1. Block diagram of monitoring system

At the base of the system's power supply block is the power management module, the CN3791 MPPT module, which is equipped with four Micro JST connectors for consumer or generator modules. The following are connected to this module:

- the 12V solar panel to one of the renewable energy connectors (Solar IN, two connectors in parallel),
- the 18650 battery and the battery itself to the port for the energy storage elements (BAT IN),
- the SX1308 step up voltage converter with the additional function of stabilizing voltage level, which can be set to a value of 5 V via a potentiometer and is connected to BAT OUT.

The main component of the system is the ESP32-DEVKIT-V1 module. Connected to it are the MQ135 gas sensor for monitoring the CO2 content in the hive, the DHT22 temperature and humidity sensor, the noise sensor based on an electret microphone, the weight sensor consisting of 50 kg

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load cells, the HX711 module and the ESP32 CAM development board for video monitoring of the hive.

The main components of the system and the connections between them are represented in Figure 2 and the schematic is shown in Figure 3.

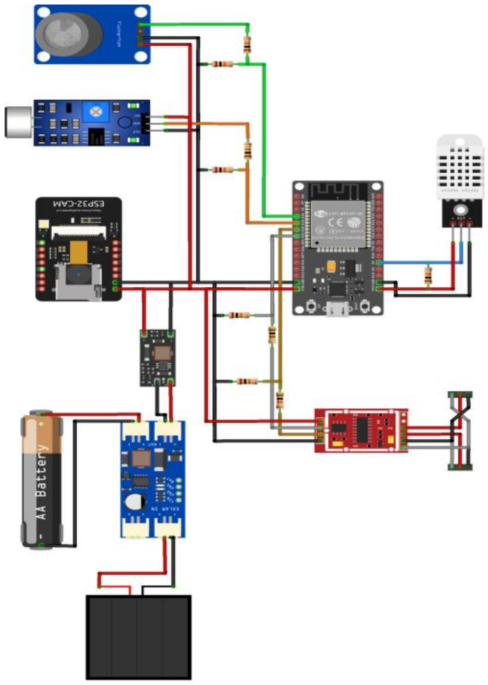

Fig. 2. Main components of the systemc

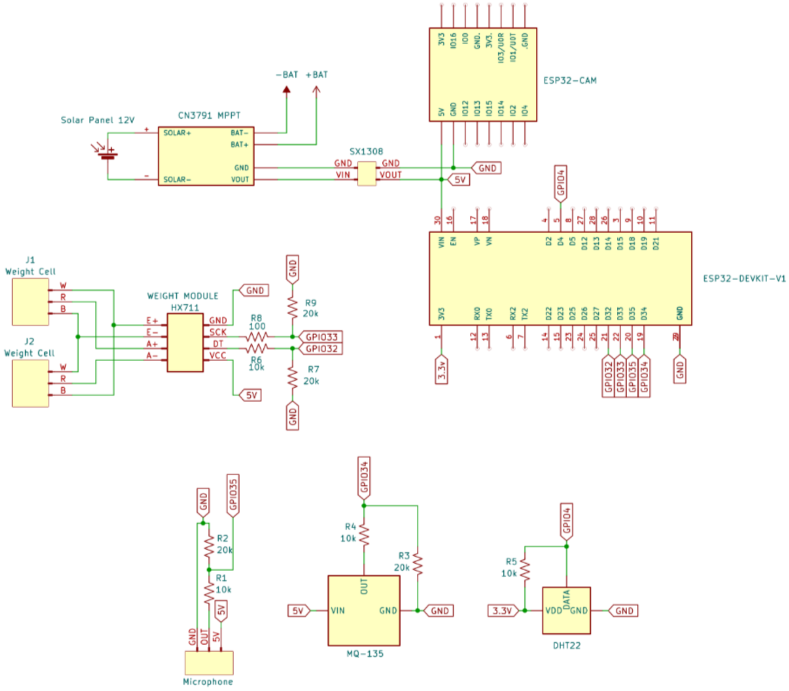

Fig. 3. Electronic schematic

A dedicated PCB was designed in Eagle PCB 9.6.2.

The layout can be observed in Figure 4 and the finished module in Figure 5.

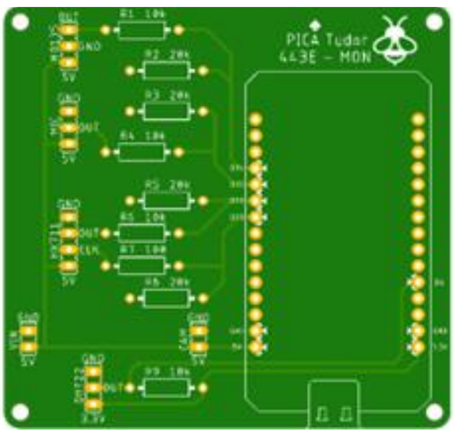

Fig. 4. .PCB layout

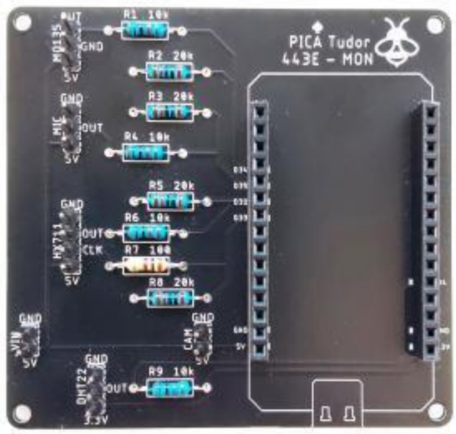

Fig. 5. Final module.

The estimated price for the system is given in table 1.

TABLE I. PRICE ESTIMATE

Component	Price (RON)
18650 + support	28.57
CN3791 MPPT	26.78
SX1308	3.56
Solar panel	20
ESP32-DEVKIT-V1	38.91
ESP32 CAM	57.32
DHT22	33.12
MQ135	11.89
Microphone	8.33
Load sensor 50kg	15.68
HX711	5.69
PCB	59.8
Resistors, wires, connectors	12.5
Others (delivery, 3D print, etc.)	70
TOTAL:	392.15

B. Software

Fig. 6. Flow chart.

After switching on the boards, a connection to a known WiFi network is established and the boards are configured. This process is repeated in a loop until a connection to an available network with a satisfactory signal is established. Next, the sensors connected to the ESP board are checked and initialized. The connection to the Firebase database services is then established using identification keys and user credentials.

This step completes the common process between the two ESP boards. On the ESP32 CAM board, the next step is to send the IP address generated by connecting to one of the configured and available WiFi networks to the database, although this address varies depending on the network set up. After sending the IP address, the ESP32 CAM module starts a loop in which it sends video images of the hive monitoring according to the requirements set by the user. This loop is repeated until the module's power supply is switched off or interrupted.

In the case of the ESP32-DEVKIT-V1 module, which is responsible for centralizing the data from the sensors and sending it to the database, the ESP enters the data acquisition loop following the identical process described above, which first checks the time of day at which the reading is taken, using the library time.h.. If it is determined that it is night, the module sends the data from the sensors to the database less frequently by an adjustable factor of 3, i.e. it sends them 3 times less frequently. The collected data is centralized and processed by the ESP32 DEVKIT-V1 module according to the functions of the libraries of the respective sensors or processed as needed and later sent to the database provided by Firebase by authenticating the module and connecting it to the service.

Following this process, the data is sent to be processed and then is made available in the online hive monitoring platform, the creation process of which is explained in the following paragraphs.

We have opted for the Firebase service from Google Inc, which offers a number of advantages. Firstly, the structure of the data management is based on the operation of a NoSQL database. This means that the data is stored in the form of object files, in this case JSON (JavaScript Object Notation) type files, which have no data mapping restrictions, are not limited to simple rows and columns and allow data to be placed in any form and relationship between them. Each node of the system is identified by a unique ID ("identifier"). Under the node's unique key (ID) you will find the list of parameters managed by the database, a list that corresponds to each sensor used in the data acquisition network. The data is sent at a specific time, converted and transmitted in Epoch Time format. Under each time point at which the data is sent, the information generated by the sensor and the time at which it was generated are written.

Through Google Firebase and the page hosting service it provides, the page is accessible to any user with the correct credentials on any device with a graphical interface and from any point of contact with the Internet network. The system for user login and registration was chosen so that it remains a query system for the project owner, where user registration cannot be performed by unauthorized users, but the possibility of registration is easy for the person responsible.

The monitoring platform was developed in the programming languages HTML, which was used for the structure of the website and the combination of its elements, JavaScript, which was used for the operating logic behind the interface and communication with the Firebse database, and CSS, which was used to create the graphical interface elements. The main user interface can be seen in figure 7.

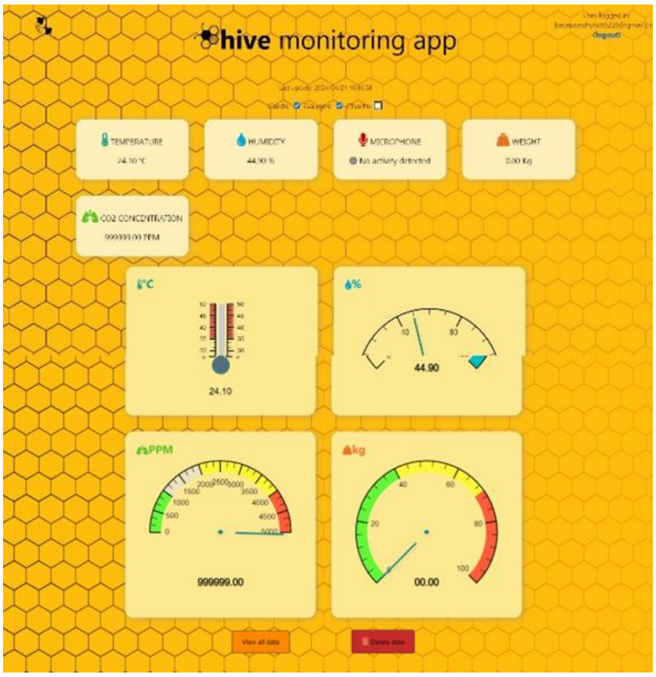

Fig. 7. User interface.

C. Testing

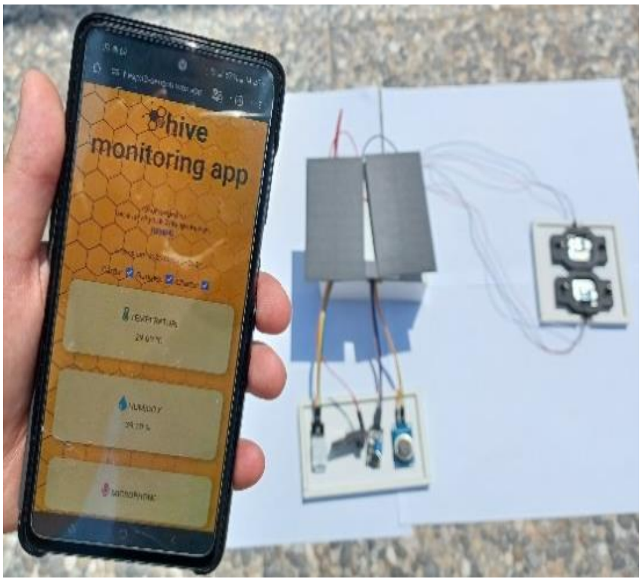

Fig. 8. Basic test setup.

The basic test setup is shown in Figure 8. The measured values were displayed on the monitoring platform. The values matched those measured with commercial sensors.

The system worked reliably throughout the test period, but although the data is transmitted correctly and the platform functions as expected, the autonomy of the system can be significantly improved. Currently, the use of low-quality components that do not meet the energy requirements of the system is a major obstacle. The problems identified, such as the use of solar modules with insufficient power and a voltage below the threshold required for the CN3791 module, underline the need to review the energy solutions.

III. CONCLUSIONS

To summarize, the system is flexible and adaptable and can be connected to a central power grid if required. This provides a temporary solution to the limitations identified during preliminary tests. Future improvements to the energy components and the use of more efficient energy sources can ensure better performance and longer autonomy.

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